

Event-Driven Video Restoration With Spiking-Convolutional Architecture

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Abstract—With high temporal resolution, high dynamic range, and low latency, event cameras have made great progress in numerous low-level vision tasks. To help restore low-quality (LQ) video sequences, most existing event-based methods usually employ convolutional neural networks (CNNs) to extract sparse event features without considering the spatial sparse distribution or the temporal relation in neighboring events. It brings about insufficient use of spatial and temporal information from events. To address this problem, we propose a new spiking-convolutional network (SC-Net) architecture to facilitate event-driven video restoration. Specifically, to properly extract the rich temporal information contained in the event data, we utilize a spiking neural network (SNN) to suit the sparse characteristics of events and capture temporal correlation in neighboring regions; to make full use of spatial consistency between events and frames, we adopt CNNs to transform sparse events as an extra brightness prior to being aware of detailed textures in video sequences. In this way, both the temporal correlation in neighboring events and the mutual spatial information between the two types of features are fully explored and exploited to accurately restore detailed textures and sharp edges. The effectiveness of the proposed network is validated in three representative video restoration tasks: deblurring, super-resolution, and deraining. Extensive experiments on synthetic and real-world benchmarks have illuminated that our method performs better than existing competing methods.

Index Terms—Convolutional neural networks (CNNs), event camera, spiking neural networks (SNNs), video restoration.

I. INTRODUCTION

VIDEO is a crucial data source for computer vision. Compared with the single image, the video contains abundant spatial and temporal information, so numerous researchers focus on the application of videos in daily life [1]. Videos recorded in real-world scenes, however, always suffer from various degradation factors, such as low resolution (LR), adverse weather, and blurring [2], [3]. These degradations

often result in detailed information loss, which seriously influences the visual experience. To address this problem, video restoration aims to obtain a visually pleasing high-quality frame sequence from low-quality (LQ) measurements. It plays a fundamental part in lots of computer vision tasks, such as tracking and video generation [4], [5], [6]. In recent years, with the wide use of convolutional neural network (CNN), researchers have achieved promising performance in numerous restoration tasks, such as video deblurring [7], [8], [9] and video super-resolution [10], [11], [12]. Although they are valid in certain scenarios, they may fall short when suffering from serious motion blur in high-speed or high dynamic scenes in complex environments. Due to the considerable loss of visual information, restoring the scene details is hardly possible [13].

Recently, a novel bioinspired sensor called an event camera has provided a new perspective for video restoration. Unlike the imaging process of traditional cameras, where frames are captured at a fixed rate, the event camera asynchronously records per-pixel brightness changes and generates a stream of events to encode the location, time, and polarity of brightness changes [14], [15], [16]. With low latency, high dynamic range, and high temporal resolution, the event camera can precisely capture the pixel variants in a very short time, offering a more detailed transition between frames. Due to its numerous advantages, events have been widely used in visual processing and analysis, including video restoration. However, they still have some limitations [17], [18]. First, the current video restoration methods [19], [20] use events as extra information without taking the association between neighboring events into account, so the high temporal resolution of events cannot be fully utilized. Second, these networks ignore the rich spatial and temporal correlation between events and frames, simply concatenating them or their feature maps [as shown in Fig. 1(b) and (c)] as the input of CNNs. However, the deployment of events is critical for video restoration since events have recorded scene illumination changes asynchronously, which can present the movement of objects in the field of view. The above issues impede the further development of event-driven video restoration research.

To tackle the limitations mentioned above, we develop a spiking-convolutional network (SC-Net) architecture to tackle with event-driven video restoration, where the spatial consistency between events and frames is extensively leveraged to recover spatial textures, and the temporal correlation in event sequences is explored to constantly capture their long-term

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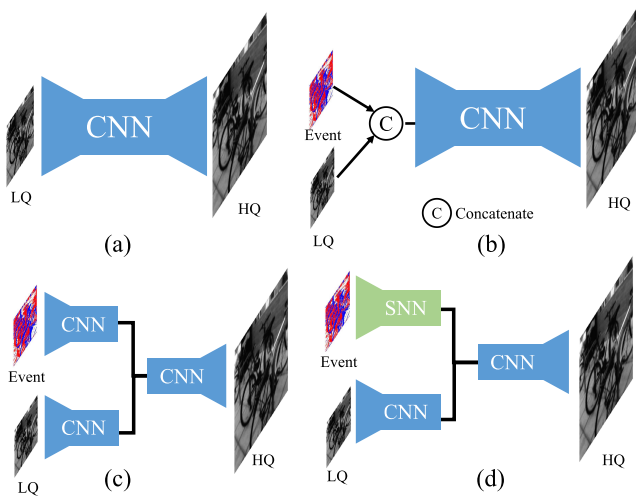


Fig. 1. Categorization of existing video restoration methods. (a) CNN-based network fed by LQ frames only. (b) Directly concatenate the events and frames along the channel dimension before feeding them into the CNN-based network. (c) Extract the feature of events and frames by CNN and fuse them. (d) Proposed hybrid network, including SNN and CNN to encode events and frames, respectively.

dependencies. Specifically, to make use of the high temporal correlation in events, the spiking neural temporal memory (SNTM) module is proposed to continually restore temporal information among various sequences of events. This module is built by adopting a spiking neural network (SNN) to suit the distribution of events and compute the long-term dependency of neighboring events to capture temporal correlation. Then, we build a frame-event spatial aggregation (FESA) module based on CNNs to fuse the two types of features by computing the nonlocal features between frames and events to capture spatial consistency. As shown in Fig. 1(d), the hybrid network deploys the SNN to process events considering their sparse nature, while frames are fed into the CNNs to extract spatial features. Compared with Fig. 1(a)–(c), our SC-Net architecture aims to continuously preserve the spatial and temporal characteristics from event sequences. Furthermore, the proposed network architecture can cope with different low-level vision tasks, including video deblurring, video super-resolution, and video deraining.

The main contributions of our work are given as follows.

- 1) We propose a unified SC-Net architecture that is able to fully explore and exploit the spatiotemporal correlations between frames and events to reconstruct high-quality frames.
- 2) To utilize high temporal resolution from events effectively, we propose an SNTM module by capturing long-term relations of event sequences in each timestamp. It transforms the brightness changes as an extra prior to aware detailed textures.
- 3) To make use of the spatial consistency, we propose an FESA module, which extracts the spatial correlation between frames and events to exploit the complementary information from them.
- 4) Extensive experiments have demonstrated the effectiveness of our approach on a range of restoration tasks,

including video deblurring, super-resolution, and deraining. The numerical results confirm the superiority of our approach.

A preliminary version of this article appears in [21]. The new contributions are mainly from four aspects: 1) we investigate the performance of SNN to deal with events and deploy SNN to cope with the sparse characteristics of events, extracting spatial and temporal information from events properly; 2) we redesign our network and find that the proposed hybrid architecture obtains better performance than our previous method STRA [21], e.g., it obtains 0.46-dB peak signal-to-noise ratio (PSNR) gains in the GoPro dataset, with almost the same parameter numbers; 3) our previous work [21] only handles video deblurring, and in this work, we extend our proposed approach into video super-resolution and deraining and compare our method with previous work [21] to illuminate the superior performance of our model; and 4) we add considerable analysis to the original version, such as the effect of the SNN, visualization of the SNN, and network parameters.

The structure of this article is given as follows. Section II provides an overview of related research on event-driven video restoration. Section III presents the SC-Net architecture that we propose. The implementation details and experimental results for three video restoration tasks are discussed in Section IV. Section V presents some limitations of our method, and Section VI offers concluding remarks on this work.

II. RELATED WORK

A. Event-Driven Video Restoration

Event cameras record intensity changes of the scene at microsecond level with slight power consumption [22] and have potential applications in video restoration, including deblurring [8], [17], super-resolution [23], [24], and reconstruction [25]. Event-based video restoration methods can be categorized into two groups, including model- and data-driven algorithms. Model-based algorithms rely on the physical event generation principle to reconstruct high-quality frames. However, due to the imperfections of physical implementation, including intrinsic noise and limited bandwidth, real events in temporal and spatial domains inevitably degrade the performance. Pan et al. [26] formulated event-based motion blurring with a double integral model. Yet, the noisy hard sampling mechanism of event cameras often introduces strong accumulated noise and loss of scene details/contrast. Jiang et al. [17] proposed a sequential formulation of event-based motion deblurring and then unfolded its optimization steps as an end-to-end deep architecture. However, event modules in these methods are too trivial to incorporate into other existing networks, making it infeasible to benefit from state-of-the-art video deblurring methods. Data-driven algorithms [23], [27] address the above limitations by utilizing neural networks and directly learn the relation of an LQ image and a sequence of sharp clear images with the aid of events. For training purposes, a synthesized dataset composed of labeled events and blurry images is commonly simulated from sharp clear video sequences. Wang et al. [19] built a synthetic dataset

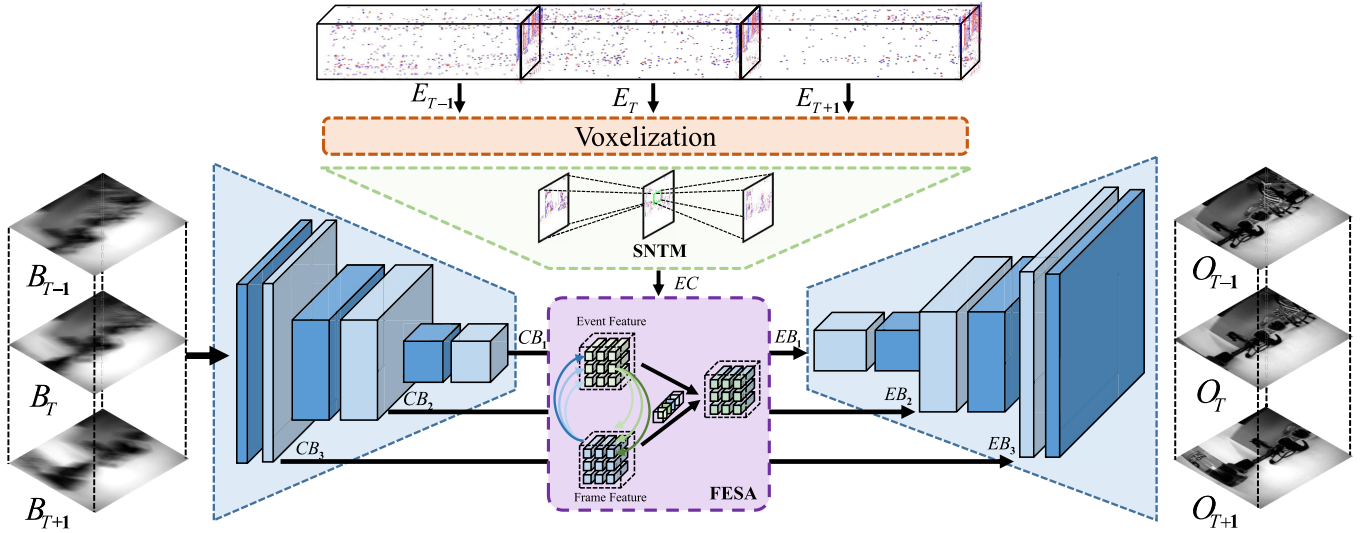


Fig. 2. Overall architecture of our proposed method. It contains two specifically designed modules: the SNTM module and the FESA. Given the triplet blurring consecutive frames $\{B_{T-1}, B_T, B_{T+1}\}$ and their corresponding event sequences $\{E_{T-1}, E_T, E_{T+1}\}$, SNTM can fuse them by computing the spatial consistency between them; FESA can capture long-term correlations of neighboring event sequences continuously in temporal order.

based on GoPro [28] to connect events, LR blurry images, and the high-resolution (HR) sharp clear images and proposed a sparse learning network to recover the high-quality images from event cameras. Compared to the abovementioned methods, our proposed approach differs in the following aspects: 1) we consider the long-range dependency of different events to restore temporal event correlation; 2) we calculate spatial consistency between events and frames with improved nonlocal operation to capture degraded contexts; and 3) our unified framework can resolve several restorations, such as deblurring, super-resolution, and deraining.

B. Spiking Neural Networks

SNN is a bioinspired network that can naturally suit the asynchronous and sparse characteristics of event streams by using biomimetic spike neurons as computing units [29], [30]. For the reproduction of this asynchronous and spike-based mechanism, multiple models of spiking neurons have been presented at various levels of abstraction. Biophysical formulations lead to accurate representations of neural dynamics, and however, their complexity limits their use in large-scale networks [31], [32]. Because it has a specific event-triggered computation characteristic that can respond to the events in a nearly latency-free way, it is naturally fit for processing events and can preserve the spatial and temporal information of events by utilizing a discretized input representation. Yao et al. [33] proposed a temporalwise attention SNN by implementing an attention mechanism in temporalwise input to filter out useless frames for event stream classification. Paredes-Vallés et al. [34] presented a methodology for optical flow estimation using convolutional SNNs based on spike-time-dependent-plasticity learning. The main limitation of these works is that they employ shallow SNN architectures because deep SNNs suffer in terms of performance. In view of these, a hybrid approach is expected to obtain the advantages of SNN and CNN.

III. METHODS

A. Event Representation

Event-driven video restoration intends to recover high-quality frames $\{I_T\}$ from LQ frames $\{B_T\}$ in one time step T and its corresponding event sequence E_T

$$E_T \triangleq \{(x_n, y_n, p_n, t_n)\}_{n \in T} \quad (1)$$

where (x_n, y_n) and t_n denote the spatial and temporal location of the n th event, respectively. $p_n \in \{+1, -1\}$ is the polarity (increase or decrease). The polarity is computed by

$$p_n = \Phi \left(\log \left(\frac{L_{xy}(n)}{L_{xy}(n-1)} \right), c \right) \quad (2)$$

where c is the intensity threshold deciding whether an event can be recorded or not and $L_{xy}(n)$ and $L_{xy}(n-1)$ represent the instantaneous brightness intensity at location (x, y) at time n and $n-1$, respectively. $\Phi(\cdot, \cdot)$ is a piecewise function

$$\Phi(x, c) = \begin{cases} -1, & x \leq -c \\ +1, & x \geq +c \end{cases} \quad (3)$$

and many deep learning-based methods are proposed to solve this problem by learning a mapping function, i.e.,

$$I = f(B, E) \quad (4)$$

where $f(\cdot, \cdot)$ is the representation from LQ frame B and events E to the corresponding high-quality frame I .

B. Network Structure

The overview of our proposed SC-Net architecture is shown in Fig. 2. First, our network feeds three LQ frames into an encoder to extract frame features. Then, to utilize the temporal information from events, the SNTM module will continuously capture temporal information between different sequences of events. Moreover, the FESA module is proposed to fuse two types of features by computing nonlocal correlations.

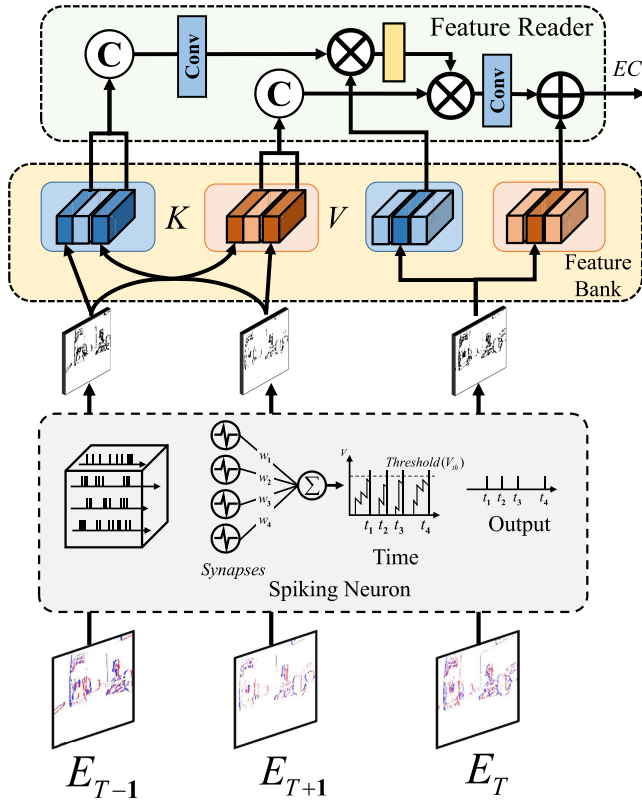


Fig. 3. Structure of SNTM module. The input event sequences are fed into SNN to extract features, and then, the “feature reader” calculates the correlation of neighboring events.

C. SNTM Module

The structure of SNTM module is shown in Fig. 3. For the event camera, the polarity of event streams denotes whether the brightness of a given pixel is increasing (ON) or decreasing (OFF). In biological brain systems, a neuron that accepts stimuli and fires a spike can be modeled using the leaky integrate-and-fire (LIF) model. It can be calculated as follows:

$$\tau_m \frac{dV^l}{dt} = -V^l + C(t) \quad (5)$$

where V^l denotes the post-neuronal membrane potential and τ_m presents the membrane potential decay time constant. The input current $C(t)$ can be calculated as the weight sum of pre-spikeings at each timestamp

$$C(t) = \sum_{i=1}^{n_l} \left(w_i \sum_k \theta_i(t - t_k) \right) \quad (6)$$

where n_l denotes the number of presynaptic weights and w_i is the synaptic weight connecting i th pre-neuron to post-neuron. $\theta_i(t - t_k)$ is a spiking event from i th pre-neuron at time t_k , which can be formulated as follows:

$$\theta_i(t - t_k) = \begin{cases} 1, & t = t_k \\ 0, & \text{otherwise} \end{cases} \quad (7)$$

where t_k is the time instant that k th spiking occurred. For training and inference, the iterative representation of LIF

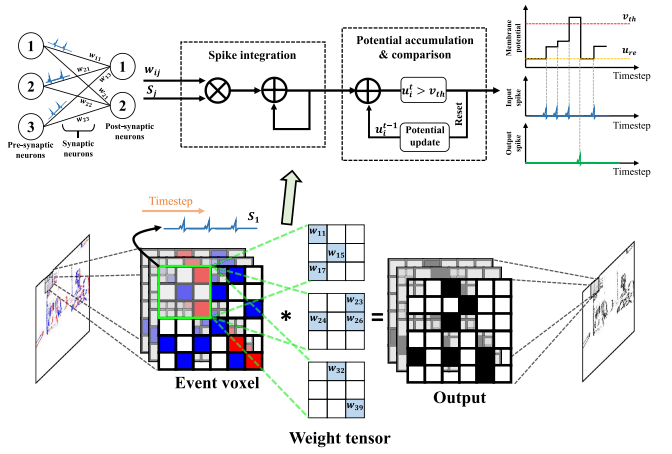


Fig. 4. Detailed SNN operations.

model can be described as

$$D[t] = f(V[t - 1], I[t]) \quad (8)$$

$$S[t] = \Theta(D[t] - V_{th}) \quad (9)$$

$$V[t] = D[t](1 - S[t]) + V_{reset}S[t] \quad (10)$$

where $I[t]$ is the input current at time-step t and $D[t]$ and $V[t]$ denote the membrane potential after neuronal dynamics and after the trigger of a spike, respectively. $\Theta(x)$ is defined by $\Theta(x) = 1$ for $x \geq 0$ and $\Theta(x) = 0$ for $x < 0$. V_{th} is the firing threshold, and $S[t]$ is the output spike, which is equal to 1 if there is a spike and 0 otherwise. V_{reset} is the reset potential. The function $f(\cdot)$ describes the neuronal dynamics and takes different forms for different spiking neuron models

$$H[t] = V[t - 1] + \frac{1}{\tau} (I[t] - (V[t - 1] - V_{reset})) \quad (11)$$

where τ represents the membrane time constant. The detailed SNN operations are shown in Fig. 4.

For training method in SNN, the total loss is evaluated after the forward propagation of all consecutive events and frames through the network, including the SNN. Due to this discontinuous and nondifferentiable neuron model, standard backpropagation algorithms cannot be applied to SNNs in their native form. We adopt the approximate gradient method [35] for backpropagating errors through SNN layers. The approximate IF gradient is computed as $(1/V_{th})$, where the threshold value accounts for the change of the spiking output with respect to the input. In the forward phase, neurons in the SNN layers accumulate the weighted sum of the spike inputs in membrane potential. If the membrane potential exceeds a threshold, a neuron emits a spike as its output and resets. The final SNN layer neurons just integrate the weighted sum of spike inputs in the output accumulator while not producing any spikes as the output. At the last time step, the integrated outputs of SNN layers propagate to the CNN layers. After the forward pass, the final loss is evaluated, followed by backpropagation of gradients through the ANN layers using standard backpropagation.

Compared with traditional convolutional layers, spike neuron can certainly maintain the spatial and temporal dynamics.

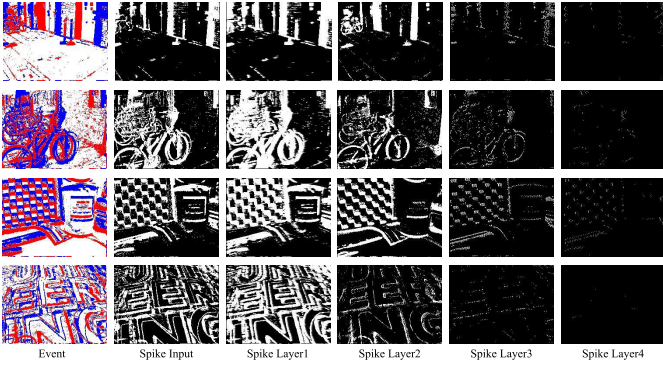


Fig. 5. Visualization of degradations in SNN at deeper layers. From left to right, the layer of SNN gets deeper, but the number of spikes vanishes drastically.

However, most of the recent works have proved that the number of spikes drastically vanishes at deeper layers, leading to serious degradation in performance [36], as shown in Fig. 5. Thus, in our spiking neural extraction, two LIF neurons are employed to extract event features.

The proposed SNTM module has a feature bank and a feature reader. For neighboring event sequence in $T - 1$ and $T + 1$, one common-used and two special-used spiking neural layers (SNLs) are deployed to obtain the key and the value. Their own SNLs are deployed on the common feature map F_m so that all keys and values of each frame will be stored by temporal order in the feature bank. The equation can be shown by

$$K_{T-1}, V_{T-1} = \text{SNL}_{T-1}(F_m) \quad (12)$$

$$K_{T+1}, V_{T+1} = \text{SNL}_{T+1}(F_m). \quad (13)$$

In the feature reader, the keys and values of neighboring events are concatenated, and the similarities between query and key to present temporal correlation with current events are calculated. In this way, the temporal correlation between the current sequence and its adjacent sequences will be completely utilized to capture long-term dependencies.

D. FESA Module

Motivated by the asymmetric integration of intrascale characteristics from different scales [37], three independent feature maps are resized to the same scale and concatenated them for fusion with event features. It can be represented as

$$\text{CB} = \text{Conv}\left(\text{Concat}\left(\text{CB}_1^\downarrow, \text{CB}_2^\downarrow, \text{CB}_3\right)\right) \quad (14)$$

in which CB_1 – CB_3 denote three independent feature maps and CB is the output of feature fusion. To maintain the same scale as the frame feature, we also utilize a downsampling layer for events. Then, the nonlocal attention operation is deployed to fuse spatial features from frames and events [38], [39] by considering the spatial correlation between all regions in each frame and their corresponding events

$$\text{Feat}_i = \frac{1}{C(E)} \sum_{\forall j} f(E_i, F_j)g(F_j) \quad (15)$$

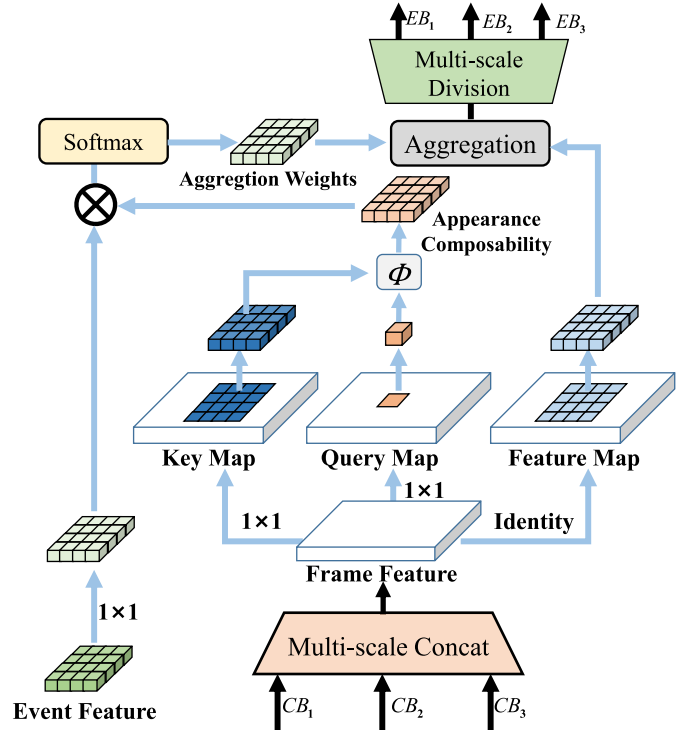


Fig. 6. Structure of FESA module. The three maps are transformed in the same scale, and then, the spatial correlation between events is calculated.

where E and F denote the event feature maps and LQ frame feature maps, respectively, and i and j represent the index of positions. The function $g(\cdot)$ computes the representation of feature maps at the position j . We set $C(x) = N$ to normalize the final result, in which N is the number of pixels in the feature map. The function $f(\cdot, \cdot)$ can be calculated as

$$f(E_i, F_j) = \theta(E_i)^T \varphi(F_j). \quad (16)$$

Finally, we deploy multiscale division by upsampling the output to three maps (EB_1 – EB_3) to connect with U-net decoder feature maps. Fig. 6 shows the structure of FESA. Unlike traditional nonlocal network, which only computes the weighted sum of all features in a single feature map, our FESA combines three frame feature maps in different scales and records correlation with corresponding event sequences.

E. Loss Function

In our work, the mean squared error (mse) is applied to train our network in an end-to-end fashion

$$\mathcal{L}_{\text{mse}} = \frac{1}{WH} \sum_{x=1}^W \sum_{y=1}^H (\hat{I}_{x,y} - I_{x,y})^2 \quad (17)$$

where $I_{x,y}$ and $\hat{I}_{x,y}$ correspond to the values of ground truth and the output of our network at location (x, y) , respectively.

IV. EXPERIMENT

A. Dataset

1) *GoPro* [28]: It contains 2103 pairs of blurry and sharp frames without corresponding events in the training process and 1111 pairs in the testing process. Following [19],

TABLE I

DETAILS OF GoPro AND HQF DATASETS. #seq, size, AND #AVG REPRESENT THE NUMBER OF DIVERSE SCENE SEQUENCES, IMAGE SIZE, AND AVERAGE NUMBER OF EVENTS WITHIN THE EXPOSURE TIME OF THE CORRESPONDING IMAGE, RESPECTIVELY

Dataset	#seq.	size	#AVG
GoPro (train)	240	180×320	164827
GoPro (test)	30	180×320	161834
HQF (train)	9	180×240	65227
HQF (test)	5	180×240	71895

we increase the frame rate by interpolating seven images between consecutive frames and then generate both events and blurry images based on the interpolated high frame-rate sequences. V2E [40] simulator is utilized to generate the corresponding event sequences. To increase the noise diversity, we set different contract thresholds for pixel-level from Gaussian distribution $N(0.18, 0.03)$ [8]. According to [27], the images are downsampled to a small size to match the real event cameras. Some details are provided in Table I, where the number of diverse scene sequences #seq, image size, and average number of events within the exposure time of the corresponding blurry image #AVG are presented.

2) *HQF* [18]: It consists of sharp ground-truth frames and corresponding events that were simultaneously captured from DAVIS240C event camera [14] that are well-exposed with little motion blur. Following [27], we synthesize motion blurs in the same way as the GoPro dataset [28]. Moreover, because HQF is captured in the real world, we also regard it as the real-world testing dataset in video SR task to demonstrate the robustness of our method in the real-world scene.

3) *Need for Speed-SR*: Following [24], we rely on synthetic datasets to train our network because there is no public dataset that contains LR frames, ground-truth HR frames, and corresponding event streams. In particular, we employ the need for speed (NFS) [41] as sources, and all frames are downsampled to 128×128 as LR frames. The corresponding event streams are generated from V2E [40], and the HR frames are produced in different scales according to upscale factors ($\times 2$ and $\times 4$). Finally, there are 3828 tuples produced from 132 video sequences. To evaluate our method, the testing dataset, which is the same as in [24], contains 841 intensity images and the related simulated event streams between two successive frames from 19 videos.

4) *NTURain*: It is proposed by Chen et al. [42]. The videos are captured by the camera with diverse motions for each clip, e.g., panning slowly with unstable movements, or mounting on a fast moving vehicle. The rain streaks are synthesized with adjustable parameters, such as raindrop size, opacity, scene depth, wind direction, and camera shutter speed. It has 24 paired rainy sequences with clear versions for training and eight pairs for testing. The corresponding events are also generated by V2E [40].

B. Implementation Details

We randomly cropped frames in 256×256 size and horizontally flipped them with a probability of 0.5 to augment the training set. The ADAM optimizer [43] is with parameter

$\beta_1 = 0.9$ and $\beta_2 = 0.999$ and trains our network in a batch size of 8. The initial learning rate is set as 10^{-4} , and it decreases by 25% every 50 epochs; the maximum training epoch is 200. The framework is implemented in Pytorch and our experiments are deployed on a single NVIDIA RTX 2080Ti GPU.

C. Evaluated Metrics

Image quality refers to visual attributes of images and focuses on the perceptual assessments of viewers. We chose PSNR, structural similarity index (SSIM), and perceptual image patch similarity (LPIPS) as our metrics.

1) *Peak Signal-to-Noise Ratio*: It is defined via the maximum pixel value and the mse between images. Given the ground-truth image I with N pixels and the reconstruction $\hat{I}$, the PSNR value can be calculated as

$$\text{PSNR} = 10 \cdot \log_{10} \left(\frac{L^2}{\frac{1}{N} \sum_{i=1}^N (I(i) - \hat{I}(i))^2} \right) \quad (18)$$

where L is equal to 255 in general cases using 8-bit representations. Since the PSNR is only related to the pixel-level mse, only caring about the differences between corresponding pixels instead of visual perception, it often leads to poor performance in representing the reconstruction quality in real scenes, where we are usually more concerned with human perceptions.

2) *Structural Similarity Index*: It is proposed for measuring the structural similarity between images based on independent comparisons in terms of luminance, contrast, and structures. For an image I with N pixels, the luminance μ_I and contrast σ_I are estimated as the mean and standard deviation of the image intensity, respectively. Also, the comparison functions on luminance, contrast, and structure, denoted as $C_l(I, \hat{I})$, $C_c(I, \hat{I})$, and $C_s(I, \hat{I})$, can be calculated as

$$C_l(I, \hat{I}) = \frac{2\mu_I\mu_{\hat{I}} + C_1}{\mu_I^2 + \mu_{\hat{I}}^2 + C_1} \quad (19)$$

$$C_c(I, \hat{I}) = \frac{2\sigma_I\sigma_{\hat{I}} + C_2}{\sigma_I^2 + \sigma_{\hat{I}}^2 + C_2} \quad (20)$$

$$C_s(I, \hat{I}) = \frac{\sigma_{I\hat{I}} + C_3}{\sigma_I\sigma_{\hat{I}} + C_3} \quad (21)$$

$$\text{SSIM}(I, \hat{I}) = [C_l(I, \hat{I})^\alpha][C_c(I, \hat{I})^\beta][C_s(I, \hat{I})^\gamma] \quad (22)$$

where α , β , and γ are control parameters for adjusting the relative importance.

3) *Perceptual Image Patch Similarity*: Zhang et al. [45] collected a large-scale perceptual similarity dataset, evaluated the LPIPS according to the difference in deep features by trained deep networks, and showed that the deep features learned by CNNs model perceptual similarity much better than measures without CNNs.

D. Video Deblurring

In this section, we compare the performance of our proposed framework with several state-of-the-art video deblurring methods, including eSL-Net [19], STRA [21], STFAN [1], RED-Net [27], MemDeblur [46], LEDVI [8], GShift-Net [47], DLEFNet [48], and EVDI [44].

TABLE II

AVERAGE PSNR, SSIM, AND LPIPS RESULTS ON THE GoPro AND HQF DATASETS. THE BEST PERFORMANCE IS HIGHLIGHTED IN BOLD AND THE SECOND-BEST PERFORMANCE IS UNDERLINED. “V” AND “E” REPRESENT THE IMAGE SEQUENCES AND EVENT SEQUENCES, RESPECTIVELY

Dataset	Input	Methods	LEDVI	GShift-Net	eSL-Net	STFAN	RED-Net	MemDeblur	EVDI	DLEFNet	STRA	Ours
GoPro	V	PSNR	20.46	27.48	21.30	26.62	27.16	27.01	24.58	27.30	26.82	26.99
	V	SSIM	0.825	0.860	0.733	0.810	0.814	0.852	0.820	0.872	0.792	0.856
	V	LPIPS	0.206	0.081	0.187	0.101	0.089	0.097	0.128	0.079	0.099	0.087
GoPro	V+E	PSNR	22.86	28.57	22.59	28.07	28.98	27.31	26.40	28.95	29.73	30.19
	V+E	SSIM	0.733	0.889	0.750	0.836	0.849	0.860	0.829	0.893	<u>0.927</u>	0.939
	V+E	LPIPS	0.156	0.075	0.161	0.086	0.077	0.093	0.104	0.074	<u>0.071</u>	0.063
HQF	V	PSNR	21.37	26.57	24.99	23.19	24.63	22.69	24.41	26.98	26.33	26.69
	V	SSIM	0.789	0.873	0.841	0.812	0.831	0.807	0.818	0.879	0.866	0.865
	V	LPIPS	0.185	0.101	0.132	0.150	0.127	0.159	0.131	0.103	0.105	0.109
HQF	V+E	PSNR	22.22	26.86	25.42	24.17	25.72	23.10	25.27	26.91	27.54	27.85
	V+E	SSIM	0.687	0.901	0.754	0.711	0.763	0.813	0.841	<u>0.907</u>	0.834	0.937
	V+E	LPIPS	0.168	0.098	0.116	0.126	0.112	0.151	0.118	0.093	<u>0.091</u>	0.087

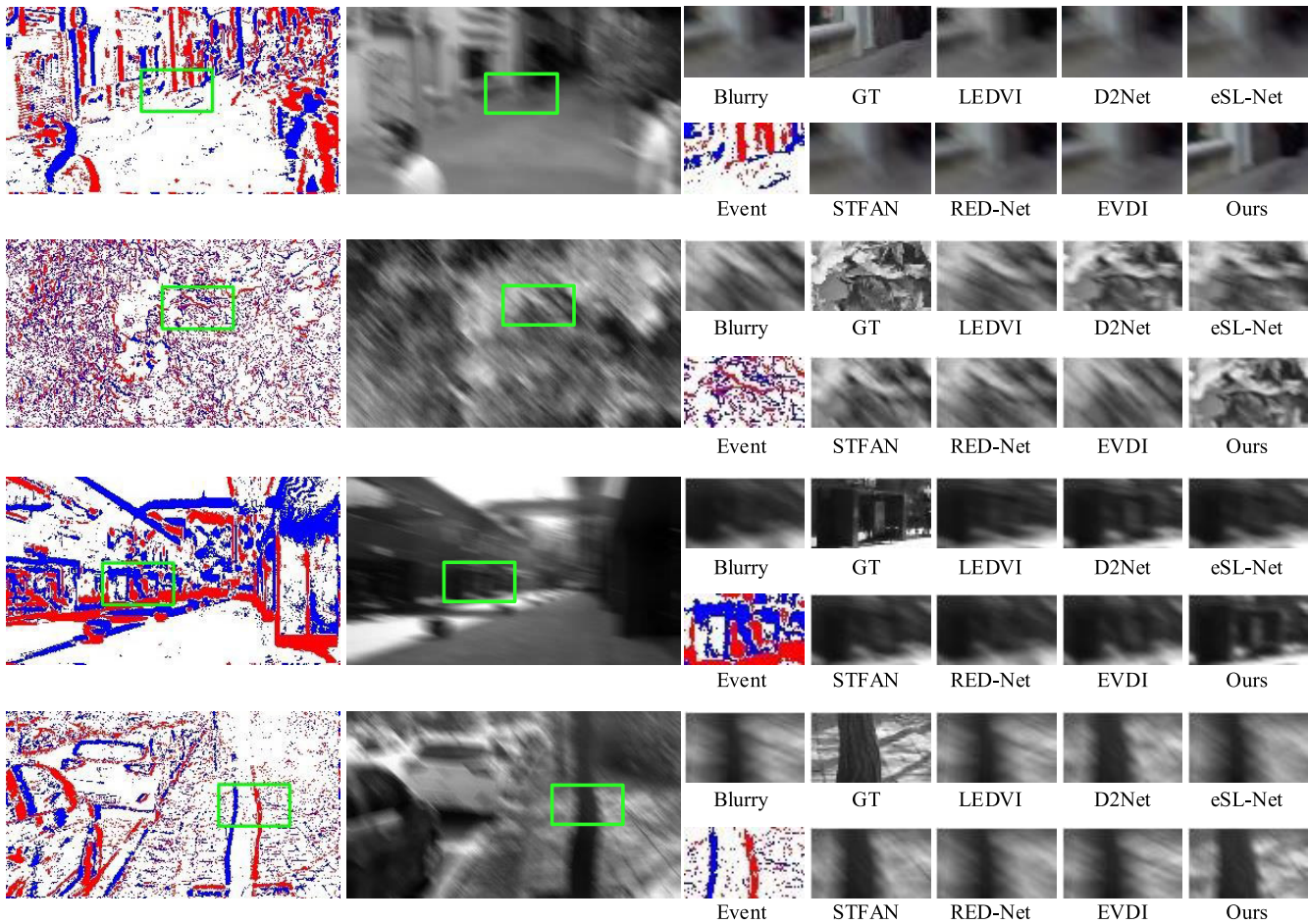


Fig. 7. Qualitative comparison of event-driven video deblurring in the GoPro dataset. “GT” is the abbreviation of “ground truth,” which means the high-quality images without any blur. Our network generates much sharper frame with clearer edges than other methods. eSL-Net [19] and EVDI [44] do not perform well without consideration of temporal correlation in events. Others contain some blurry residual.

Table II has shown the performance in GoPro and HQF datasets. Note that when the inputs are videos and events (“V + E”), we do not change the original structure of video-based network, use a CNN backbone to extract features of events, and then employ an attention module to feed them into the main stream. It is obvious that when we feed both event streams and frames into these networks, most of them achieve better PSNR, SSIM, and LPIPS values than feeding videos

only. For example, with the guidance of events, RED-Net [27] and STFAN [1] achieve 1.82- and 1.45-dB gains over video conditions without events in the GoPro dataset, respectively. The gains on HQF dataset become even larger. Moreover, the experiments have shown the superior performance of the proposed framework in all metrics than other competing methods, with 2.88-dB improvement than MemDeblur [46] in terms of PSNR. It is because our method can effectively make full use

TABLE III

QUANTITATIVE RESULTS (PSNR, SSIM, AND LPIPS) OF $SR \times 2$ AND $SR \times 4$ RESULTS ON THE SYNTHETIC TEST DATASET. THE BEST PERFORMANCE IS HIGHLIGHTED IN BOLD AND THE SECOND-BEST PERFORMANCE IS UNDERLINED. FOR THE INPUT, “V” AND “E” REPRESENT THAT THE INPUT ARE THE IMAGE SEQUENCES AND EVENT SEQUENCES, RESPECTIVELY

Scale	Input	Methods	BSRDM	eSL-Net	EvInt	EVSR	LBNet	MRVSR	SLS	TTVSR	STRA	Ours	Ideal value
$SR \times 2$	V	PSNR	34.33	32.84	28.09	28.84	29.43	34.67	33.44	32.65	34.47	34.57	$+\infty$
	V	SSIM	0.926	0.941	0.859	0.860	0.883	0.925	0.919	0.908	0.948	0.949	1.00
	V	LPIPS	0.047	0.043	0.086	0.080	0.075	0.046	0.051	0.055	0.038	0.042	0
$SR \times 2$	V+E	PSNR	35.03	34.63	30.07	30.10	32.71	<u>35.92</u>	35.29	34.86	35.55	36.01	$+\infty$
	V+E	SSIM	0.954	0.926	0.912	0.914	0.943	<u>0.955</u>	0.952	0.950	0.960	<u>0.957</u>	1.00
	V+E	LPIPS	0.038	0.046	0.073	0.051	0.049	0.035	0.037	0.036	<u>0.033</u>	0.030	0
$SR \times 4$	V	PSNR	30.42	29.71	21.84	26.07	29.64	29.99	30.59	23.45	30.85	30.67	$+\infty$
	V	SSIM	0.919	0.912	0.851	0.816	0.915	0.914	0.913	0.887	0.920	0.921	1.00
	V	LPIPS	0.061	0.060	0.143	0.108	0.071	0.055	0.064	0.142	0.057	0.053	0
$SR \times 4$	V+E	PSNR	31.01	30.57	25.43	29.46	31.16	30.85	<u>31.20</u>	26.76	31.12	31.56	$+\infty$
	V+E	SSIM	<u>0.925</u>	0.920	0.847	0.898	0.919	0.921	0.922	0.879	0.924	0.926	1.00
	V+E	LPIPS	0.061	<u>0.051</u>	0.116	0.060	0.062	0.058	0.058	0.091	0.054	0.044	0

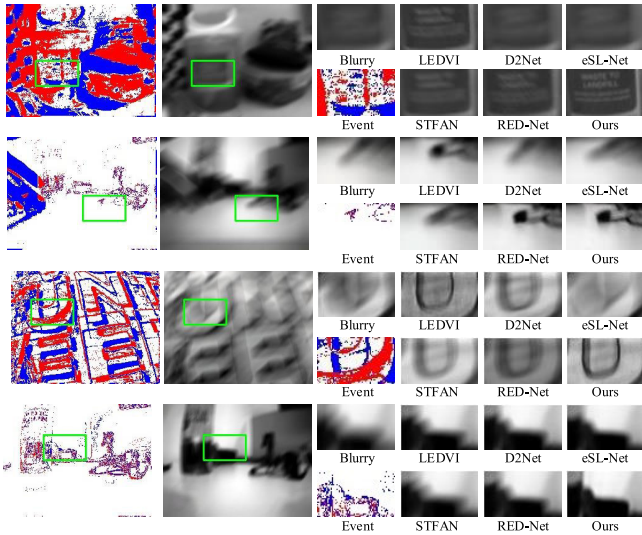


Fig. 8. Qualitative comparison of event-driven video deblurring in the HQF dataset, where the third row is the outdoor image and others are indoor images. The first column is the corresponding events to the blurry image in the second column.

of the high temporal information from events to capture the temporal relation between adjacent events as an extra prior to reconstructing sharp edges.

The performance gains over other state-of-the-art methods are consistent with the visual results. Figs. 7 and 8 present deblurring results of our method and other comparing methods in HQF and GoPro datasets, respectively. Note that “GT” is the abbreviation of “ground truth,” which means the high-quality images without any blur. Visual quality comparisons show that high-quality video deblurring with more structural details can be achieved with the right combination of frames and events. For example, in the third row of Fig. 8, the blurry edges of letter “U” is removed better by our proposed method. Other networks, such as LEDVI [8] and eSL-Net [19], however, fail to recover clear edges. Moreover, in the fourth row of Fig. 7, it is observed that in the course of blurring process, various undesired artifacts are produced in the tree’s edges. All the compared methods cannot tackle this case. However, the proposed framework can produce better visual results, more similar to the ground truth. Although the

LQ frames suffer from significant blur, events can also record the clear moving pattern of the scene, which can guide motion deblurring.

E. Video Super-Resolution

Our method is compared with several state-of-the-art methods, including BSRDM [49], STRA [21], eSL-Net [19], EvInt [24], EVSR [23], LBNet [50], MRVSR [51], SLS [52], and TTVSR [53].

1) *Synthetic Video SR*: The PSNR, SSIM, and LPIPS results of $SR \times 2$ and $SR \times 4$ are presented in Table III. From it, we have some observations. First, most of video-based SR networks can obtain obvious improvements with the help of events. Take EVSR [23] as an example, and it achieves 1.26- and 0.86-dB improvements in terms of PSNR in $SR \times 2$ and $SR \times 4$, respectively. Other competing methods also obtain similar benefits from events. This is because events can provide extra prior about high-frequency details in scenes, which contribute to the final SR results. Second, as expected, our method gets promising results in both $SR \times 2$ and $SR \times 4$ tasks in the “V + E” condition. Specifically, it achieves 36.01 dB in PSNR values in $SR \times 2$, getting the highest performance among all methods. The average gains of ours over STRA are 0.46 dB in PSNR and 0.03 in LPIPS. When compared with the other methods, our method is far ahead.

Our method not only outperforms all the above comparative methods in quantitative evaluations but also produces visually pleasing results among them. In Fig. 9, we show some visual comparisons on scales $\times 4$ in the NFS dataset. We can see that most of the state-of-the-art methods lost many texture details in the LR frame, even though MRVSR can recover partial regions. In the third row of Fig. 9, it is obvious that the event has captured the clear edges of objects in the scene, which is beneficial to recovering detailed textures from LR. However, most compared networks cannot fully utilize events to recover the lost information. On the contrary, with the proper usage of spatial and temporal correlation between events and frames, our method can obtain promising visual results with clearer structures.

2) *Real-World Video SR*: To further illuminate the performance of our proposed method, we super-resolve the

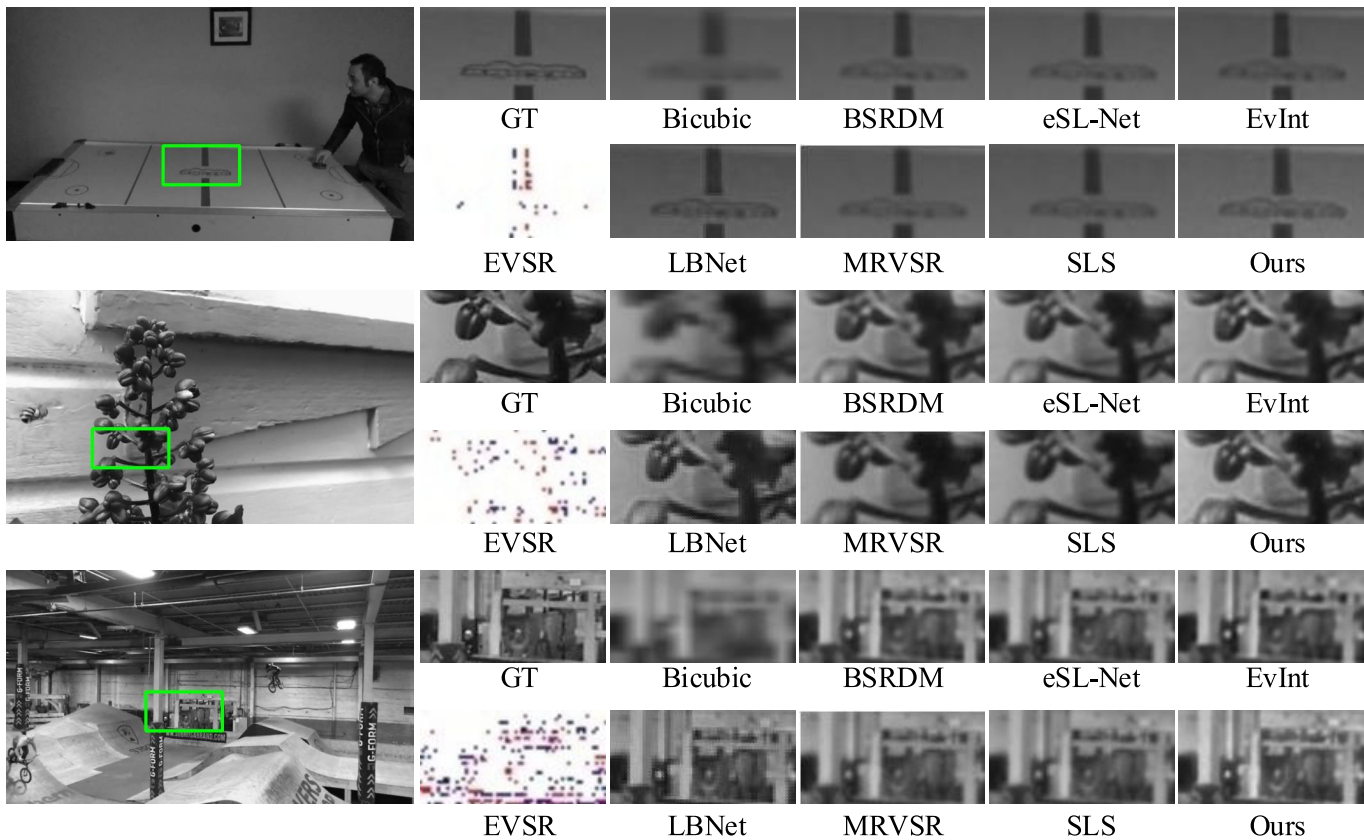


Fig. 9. Qualitative comparison of event-driven video SR $\times 4$ in the NFS dataset. Zoom in for a better view. “GT” is the abbreviation of “ground truth.” The sharpness of the edges on the objects and textures restored by our method is the best.

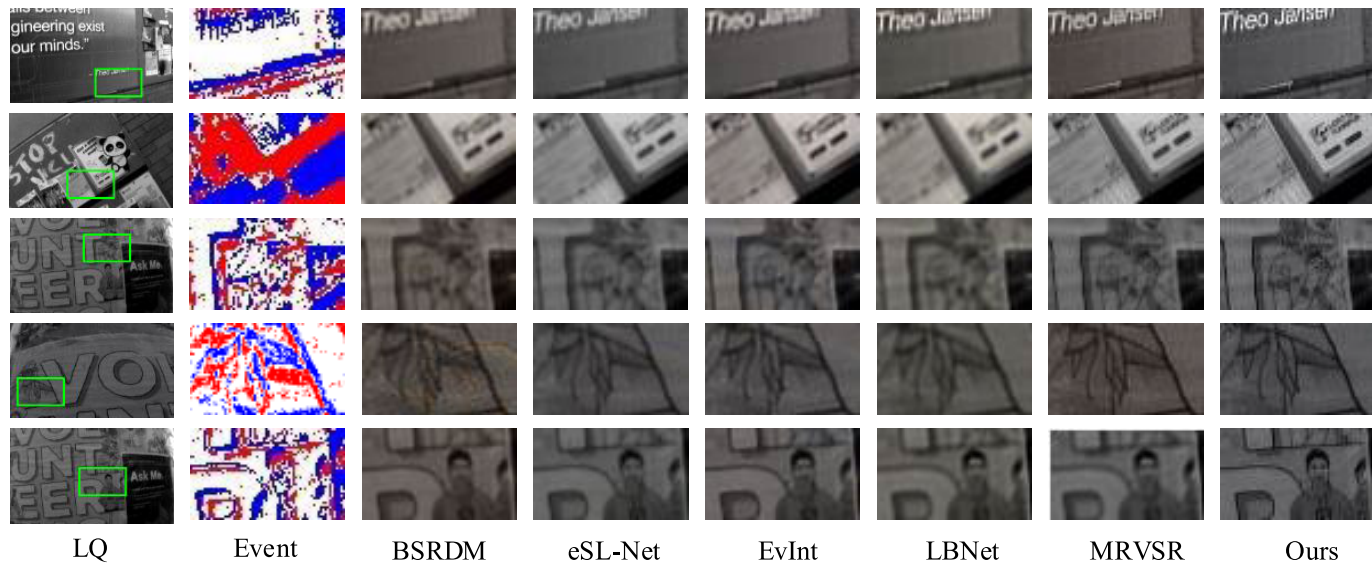


Fig. 10. Qualitative comparison of event-driven video SR $\times 4$ in the HQF dataset. The first column shows the input LR frames, and other columns show the corresponding events and SR results by different methods. Zoom in for a better view.

real-world frames with corresponding events in the HQF dataset. The results are shown in Fig. 10. For the letters “Theo” the first row, both BSRDM and eSL-Net contain some unwanted and blurred artifacts, but our method generates sharper and clearer textures. Moreover, in the last row in Fig. 10, all of the compared methods cannot reconstruct the

face of the “person” in the poster, suffering from serious texture degradation. However, our network can still produce better qualitative results. These comparisons have validated the superior performance of spatiotemporal relation-aware mechanism, allowing our method to maintain robustness in different unknown degradation conditions.

TABLE IV

QUANTITATIVE RESULTS (PSNR, SSIM, AND LPIPS) OF RESULTS ON THE NTURAIN DATASET. THE BEST PERFORMANCE IS HIGHLIGHTED IN BOLD AND THE SECOND-BEST PERFORMANCE IS UNDERLINED. FOR THE INPUT, “V” AND “E” REPRESENT THAT THE INPUT ARE THE IMAGE SEQUENCES AND EVENT SEQUENCES, RESPECTIVELY

Input	Methods	DIP	SPAC	MCSC	D3R-Net	RSRV	TCLMC	J4R-Net	VRGNet	STRA	Ours	Ideal value
V	PSNR	32.28	28.31	33.30	30.61	28.45	30.49	32.20	33.47	32.30	33.82	$+\infty$
V	SSIM	0.944	0.908	0.956	0.898	0.894	0.913	0.949	0.958	0.946	0.961	1.00
V	LPIPS	0.038	0.066	0.037	0.063	0.082	0.058	0.053	0.046	0.047	0.031	0
V+E	PSNR	34.92	30.34	33.49	32.86	30.40	31.45	32.55	33.83	<u>35.01</u>	35.29	$+\infty$
V+E	SSIM	0.967	0.913	0.959	0.950	0.919	0.935	0.949	0.958	<u>0.967</u>	0.968	1.00
V+E	LPIPS	0.016	0.066	<u>0.014</u>	0.018	0.041	0.058	0.021	0.022	0.015	0.013	0

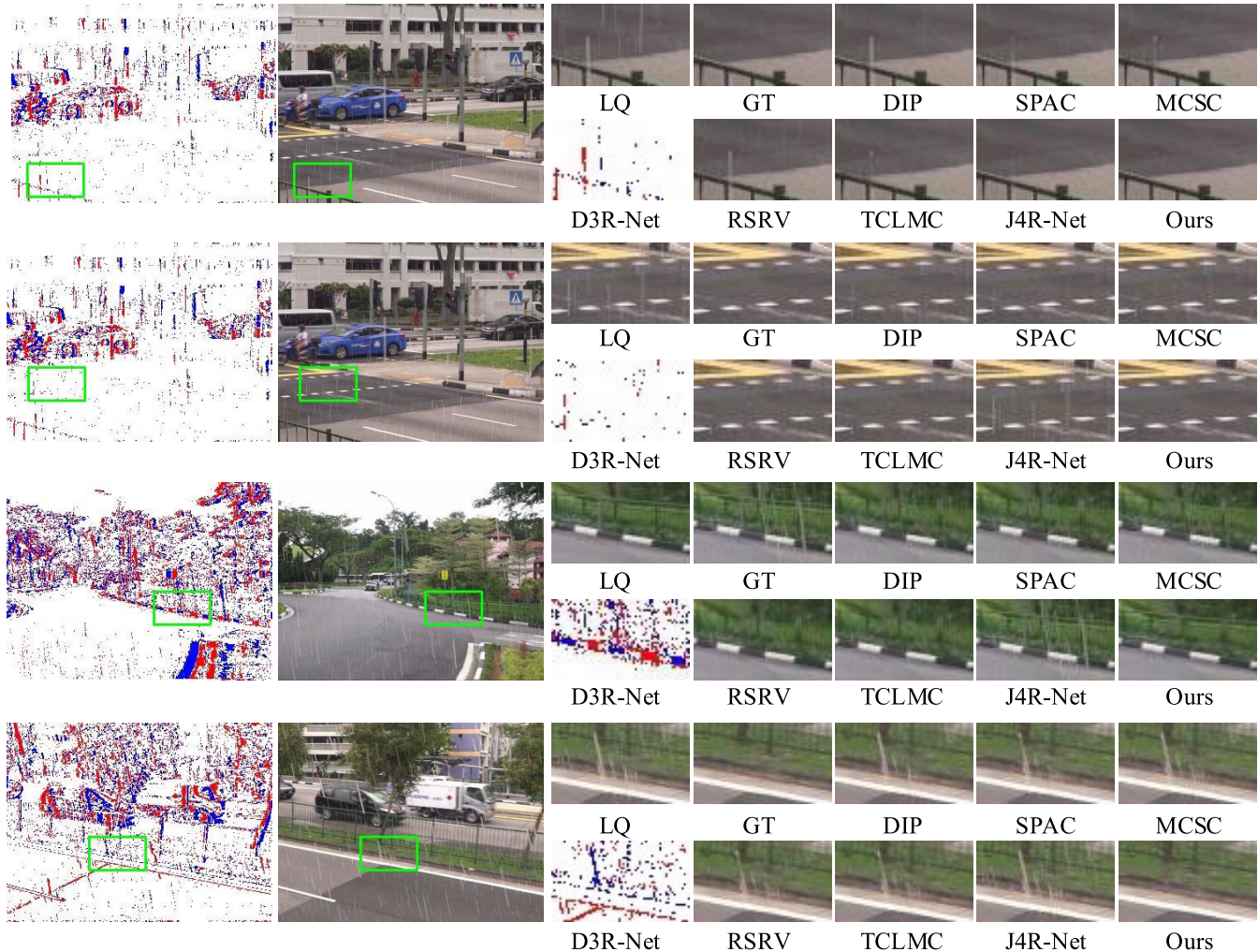


Fig. 11. Visual results of event-driven video deraining in the NTURain dataset. The first column shows the events, while the second column shows the input rainy frames. The remaining rain streaks and artifacts are denoted with green box. Zoom in for a better view.

F. Video Deraining

We compare our method with DIP [54], STRA [21], SPAC [42], MCSC [55], D3R-Net [56], RSRV [57], TCLMC [58], J4R-Net [59], and VRGNet [60].

For video deraining task, Table IV has shown that our proposed method significantly outperforms the state-of-the-art methods by consistently achieving better performance in all metrics. Compared with the second-best algorithm STRA, our method obtains a performance gain of 0.28 dB in PSNR and 0.01 in SSIM. In addition, our method gets considerable gains over other methods, i.e., 1.80 dB over MCSC [55], 2.43 dB over D3R-Net [56], and 2.74 dB over J4R-Net [59].

Clearly, when adding events into video-based networks, all of them obtain varying degrees of gains, which illuminates the benefit of guidance from events for video deraining.

Moreover, some visual comparisons on NTURain dataset are shown in Fig. 11. We have several observations.

- 1) The event camera can not only record the movement of objects in the field of view but also detect the moving streaks in the rainy day. Take the last row as an example, and the events have shown the sharp edges of street as well as the tiny rain streaks in a small region. It means that the high temporal resolution of events is beneficial to being aware of the rain streaks.

TABLE V
QUANTITATIVE ANALYSIS OF SNTM AND FESA
ON THE GoPro AND HQF DATASETS

SNN	SNTM	FESA	GoPro	HQF
0	✗	✗	27.81	25.03
0	✓	✗	29.04	25.94
0	✗	✓	28.50	25.38
0	✓	✓	29.73	27.54
1	✓	✓	29.81	27.76
2	✓	✓	30.19	27.85
3	✓	✓	28.31	26.57
4	✓	✓	28.02	26.23
5	✓	✓	27.89	26.04

TABLE VI
COMPARISON OF DIFFERENT BACKBONES ON PARAMETER NUMBERS,
FLOPS, AND THROUGHPUT. “*” MEANS THAT SNN IS UTILIZED TO
REPLACE THE TEM (TEMPORAL MEMORY BLOCK) IN STRA

Methods	# Params	FLOPS	Throughput (images/s)	PSNR
STRA	2.36M	1.10G	22.54	29.73
STRA*	2.01M	0.87G	23.18	27.12

TABLE VII
PSNR VALUES ON DIFFERENT PARAMETER SETTINGS

Channel # \ Res-unit #	$R = 4$	$R = 6$	$R = 8$	$R = 10$	$R = 12$
$C = 16$	27.99	28.77	28.98	29.12	29.36
$C = 24$	28.10	29.73	29.21	29.47	29.68
$C = 32$	28.31	29.57	30.19	30.21	30.24
$C = 48$	28.54	29.64	30.19	30.23	30.25
$C = 64$	28.67	29.72	30.22	30.24	30.25

- Although the rain streaks can be detected and removed by TCLMC and DIP, they cannot recover the missing details properly.
- There are more or less noticeable streaks generated by SPAC and MCSC, while our network can recover global texture information. In short, the rain-free frames recovered by ours achieve better performance qualitatively and quantitatively.

G. Ablation Study

1) *Benefit of Events*: The effect of events for video-based restoration is shown in Tables II–IV. We test all compared methods in both with- and without-event conditions. As we can see, the event-driven methods (“V + E”) have better performance than video-based methods, which means that events can actually bring noticeable gains in different restoration tasks. Moreover, the proposed framework can also achieve superior performance than all other networks with the guidance of events.

2) *FESA and SNTM*: We use the GoPro dataset to validate the significance of the SNTM module, and there is a significant quantitative performance gap in the first two rows of Table V. It demonstrates that due to the aid of temporal correlation between neighboring events, the deblurring performance is effectively improved with gains of 0.91 dB in the HQF dataset and 1.23 dB in the GoPro dataset in terms of PSNR.

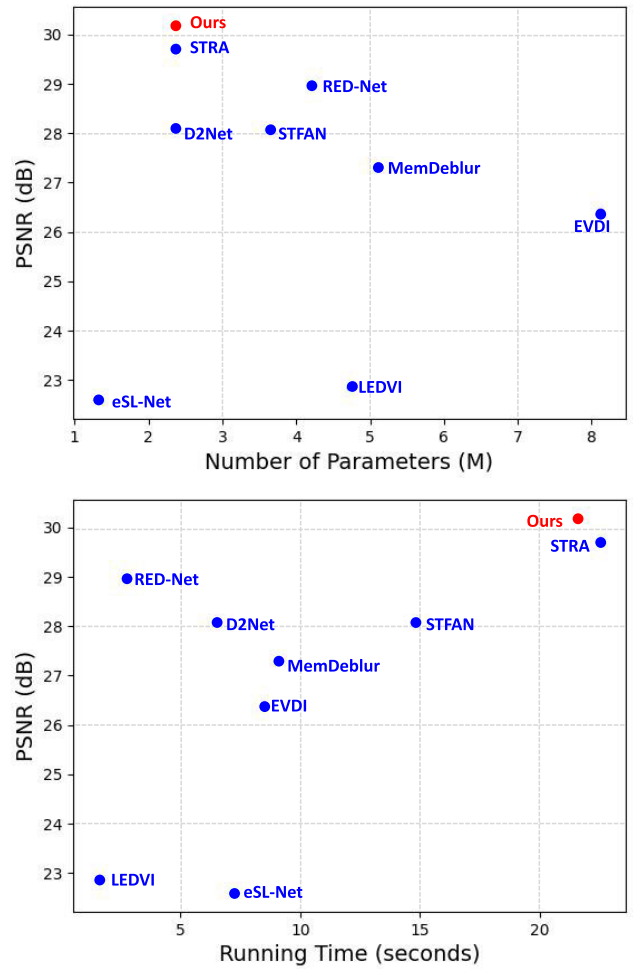


Fig. 12. Comparison of running time and parameter numbers.

TABLE VIII
ABLATION STUDY ABOUT LOSS FUNCTIONS ON
DEBLURRING IN THE GoPro DATASET

Loss function	$\mathcal{L}_1$	$\mathcal{L}_{MSE}$	$\mathcal{L}_{SSIM}$
PSNR↑	29.97	30.19	29.85
SSIM↑	0.930	0.939	0.927
LPIPS↓	0.077	0.063	0.083

The same procedure is used to examine the impact of the FESA module, and the results are shown in the first and third rows of Table V. Clearly, the usage of FESA leads to better performance, with gains up to 0.69 and 0.35 dB on GoPro and HQF datasets, respectively. Most importantly, we outperform inserting just one block in terms of deblurring performance when both SNTM and FESA are included in our network.

3) *Effect of SNN*: We also evaluate the effect of SNN, and the results are shown in the last five rows in Table V. When adding two layers of SNN, they bring 0.46- and 0.31-dB PSNR values in GoPro and HQF datasets, getting the best performance among other settings. Interestingly, if there are three SNN layers in our network, our method gets obvious degradation of performance, with dropping by 1.88 dB in terms of PSNR. It also illuminates the degradation of SNN in deeper layers, as shown in Fig. 5.

We also show FLOP changes by using SNN to replace the temporal memory block (TEM) in STRA and make a

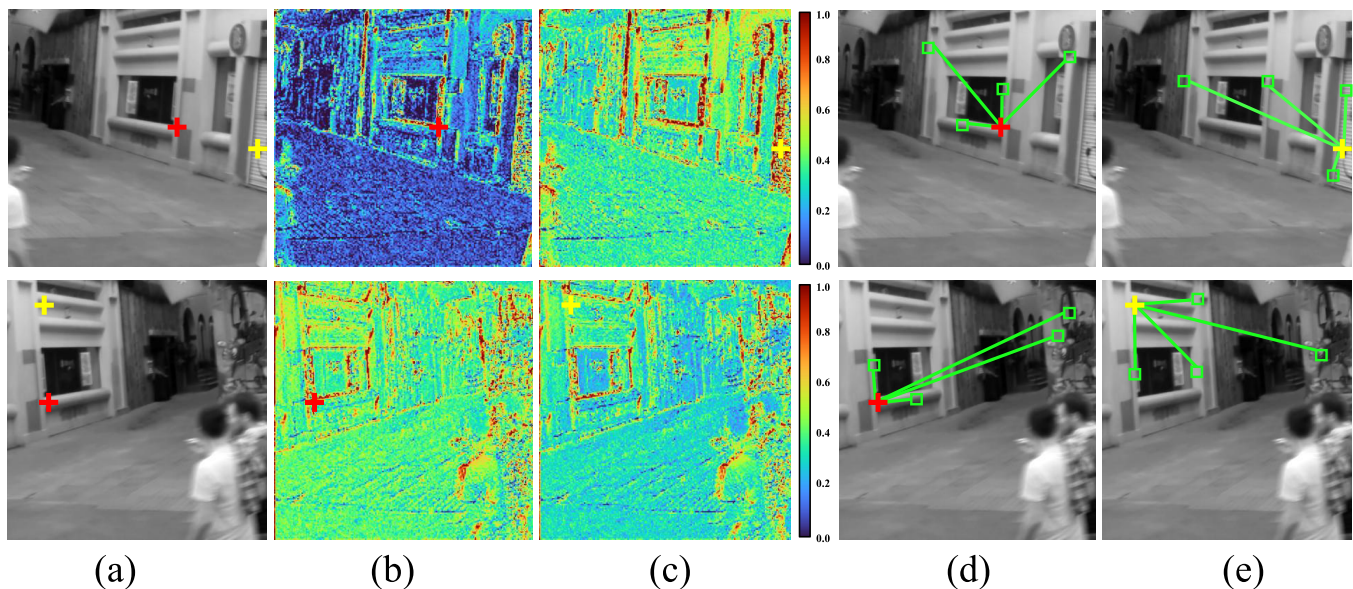


Fig. 13. Visualizations of similarity matrixes in the FESA. (a) Original blurry images. (b) and (c) Heat maps are used to display the similarity matrixes of the query points. The regions of query points are identified by red and yellow crosses. We give some highly linked neighbors in (d) and (e) as examples (marked by green boxes).

comparison with the original STRA. The results are shown in Table VI. By using SNN to replace TEM, our method obtains comparable FLOPS and running time, but the performance drops largely, which demonstrates the necessity of TEM. Due to the limitation of shallow SNN architectures, our proposed SNTM module is a hybrid approach for constructing deep network architectures to explore the temporal correlation and capture their long-term dependencies, obtaining the benefits of both SNNs and CNNs.

4) *Effect of the ResNet Depth*: We also conduct analysis about the influence of hyperparameters, including the number of channels C and res-units in our network R . Note that we set $C \in (16, 24, 32, 48, 64)$ and $R \in (4, 6, 8, 10, 12)$ in comparison. The results are shown in Table VII. Clearly, our network yields better results as the number of channels increases. Specifically, When $C = 64$ and $R = 12$, our method gets 30.25 dB PSNR values, achieving PSNR gains 0.06 dB than our original settings, but it requires lots of memory assumption and this model has to be trained for more time to obtain good performance. To achieve a good trade-off between efficiency and performance, we set $C = 32$ and $R = 8$ as the default setting.

5) *Parameters and Running Time Analysis*: To compare the running time, we evaluate different methods by processing 200 frames on an RTX 2080Ti GPU and record the average time. Results are shown in Fig. 12. With reasonable storage consumption, our method has a comparable running time and provides promising video restoration performance. Note that SNN can improve energy efficiency over time because it is only ever active when it receives or emits spikes, while CNNs, on the other hand, operate with all units active regardless of real-valued input or output values. In addition, the mathematical dot-product operation is lowered by the fact that an SNN's inputs can only be either 1 or 0. Thus, we deploy SNN by adding less than 0.01 M parameters. It is because SNN

utilizes the intrinsic sparsity of spatiotemporal events and provides asynchronous computations. Its mechanism naturally encapsulates the asynchronous processing capability, leading to energy-efficient computation.

6) *Effects of Different Loss Functions*: We conduct ablation study on different loss functions, including $\mathcal{L}_{mse}$, $\mathcal{L}_1$, and $\mathcal{L}_{SSIM}$. Note that Ren et al. [61] proposed $\mathcal{L}_{SSIM}$ and shows that it has some benefits for low-level computer vision tasks. The results are shown in Table VIII. Obviously, using $\mathcal{L}_{mse}$ can obtain the best performance. $\mathcal{L}_{mse}$ is chosen as a loss function to train our network to get better results.

H. Visual Analysis

To demonstrate the operation of spatial consistency between frames and events, various attention maps are shown in Fig. 13. Note that the results are normalized and presented in the form of heat maps, and the lighter color indicates higher importance. The location of two query patches is labeled by red and yellow boxes, and some regions with the highest weights are labeled by green boxes in Fig. 13(b) and (c). Take the first row as an example, and the lighter color in Fig. 13(b) always appears in the edge of objects, while in Fig. 13(c), it appears in the surface of walls. Clearly, the heat maps of red cross and yellow cross have great differences. The similar brightness actually has more influence on the query patches, which illuminates the effectiveness of our FESA module to present long-term spatial correlations in particular patches.

For the SNTM module, we present some similarity matrixes with neighbor events in the same way. Some highly correlated neighbors are also labeled by green boxes. In the first row of Fig. 14, the patches of red cross have no events, and its higher related regions also appear in marginal areas. In contrast, the higher related regions of the patches in the yellow cross always have more events. For event patches, the regions with frequent brightness changes are more significant, which

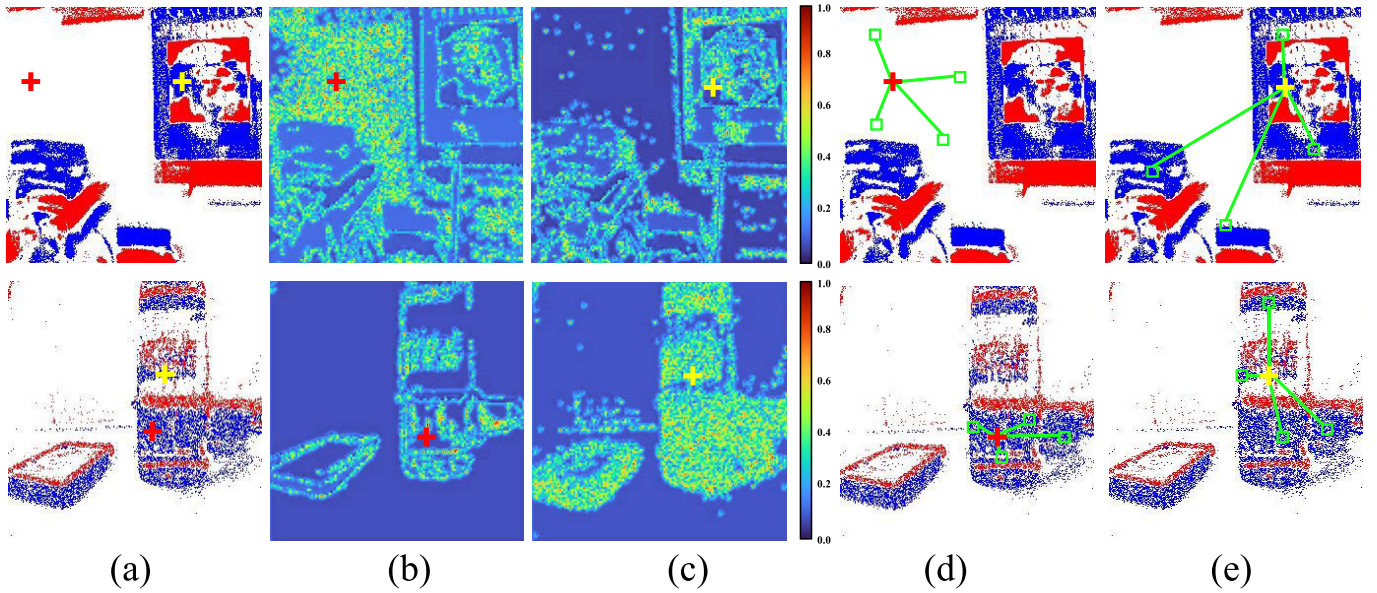


Fig. 14. Visualizations of similarity matrices in the SNTM. (a) Event visualization. (b) and (c) Heat maps are used to display the similarity matrices of the query points. The regions of query points are identified by red and yellow crosses. We give some highly linked neighbors in (d) and (e) as examples (marked by green boxes).

demonstrates how the temporal relationship between adjacent event sequences was constructed.

V. LIMITATIONS

While our method achieves the state-of-the-art performance, it still fails in some special scenarios. One limitation of the proposed method is that the network needs to be retrained if we aim to further increase the frame rate. We can solve it by applying an additional interpolation network recursively between pairs of restored sharp frames. Further research will be devoted to arbitrary frame-rate video reconstruction. For computational requirements, an approach of interest would be to perform a distributed edge-cloud implementation where the SNN blocks and CNN blocks are administered on the edge device and the cloud, respectively. This would lead to high energy benefits on edge devices, which are limited by resource constraints while not compromising on algorithmic performance.

VI. CONCLUSION

In this article, we propose a novel spiking-convolutional architecture for event-driven video restoration. By bringing in advantages of SNN and CNN to suit the characteristics of events and frames, the proposed network is capable of restoring videos suffering from various degradations. To model the long-term dependencies in events, the SNTM module continuously maintains the temporal correlation between adjacent event sequences to obtain better performance. Moreover, the FESA module calculates spatial consistency by enhancing nonlocal operations to utilize spatial contexts from them. Consequently, the framework learns to reconstruct high-quality regions depending on adequate temporal and spatial information provided by events. We evaluate the proposed method on numerous synthetic and real-world datasets, which clearly

shows that our method outperforms the state-of-the-art methods for several video restoration tasks.

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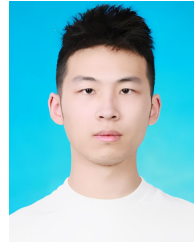
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